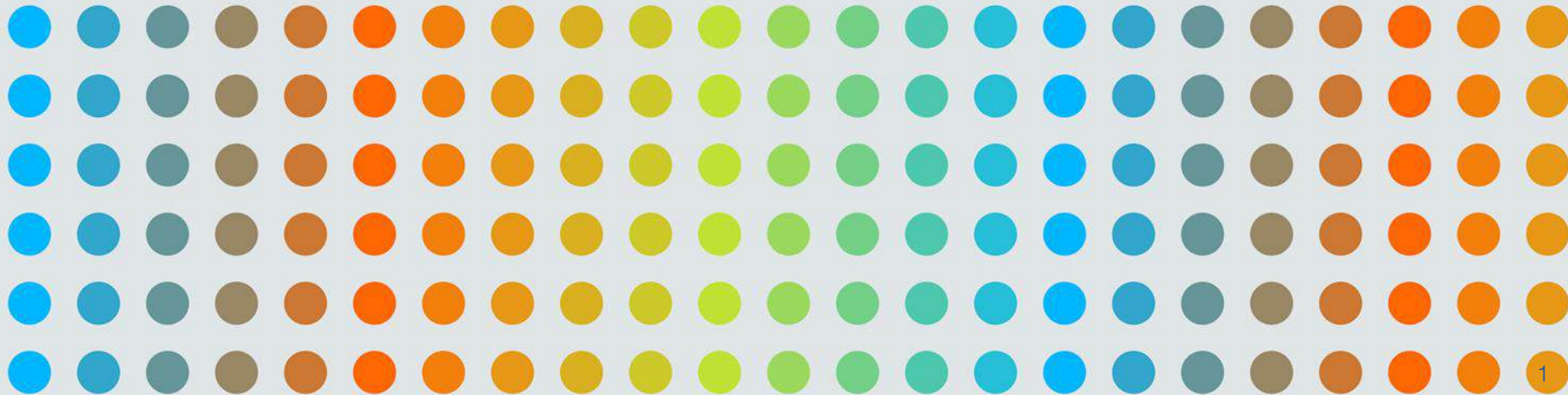




An introduction to data analysis

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Founded in 2012, the Open Data Institute (ODI) is an international, independent and not-for-profit organisation based in London, UK.



Sir Nigel Shadbolt

Executive Chair and Co-founder of the ODI



Sir Tim Berners-Lee

President and Co-founder of the ODI

Our mission & Vision

We work with companies and governments to build an open, trustworthy data ecosystem.

We want people, organisations and communities to use data to make better decisions and be protected from any harmful impacts.



Today's Aim

Enable you to apply machine learning and AI techniques to data and discover how ethical frameworks can help you avoid teaching your machines bad habits.



Introductions

Name?

What you do?

What brings you to the training today?

In chat window:

Put a random time between 4 and 8 minutes, e.g. 5m43s

Pick a colour from red, blue, green and yellow.

Schedule

09:40 - Exploratory data analysis and descriptive statistics

10:45 - Break

11:00 – Inferential statistics

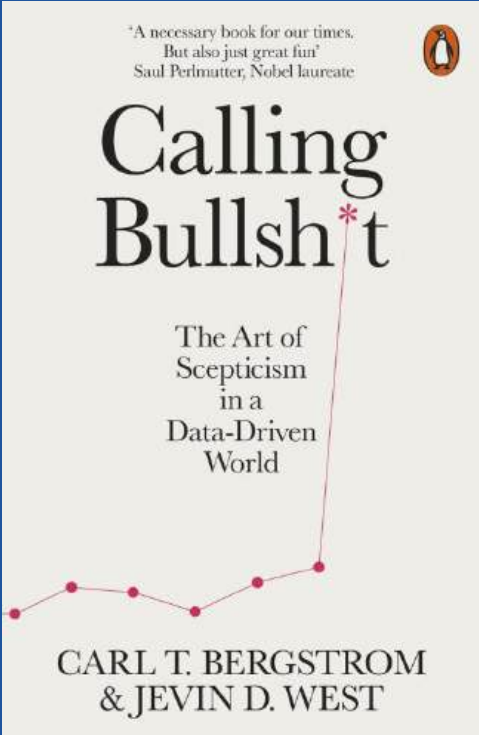
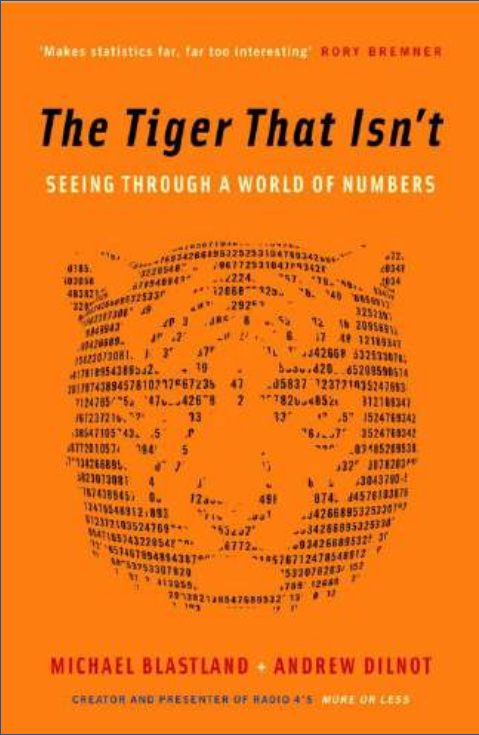
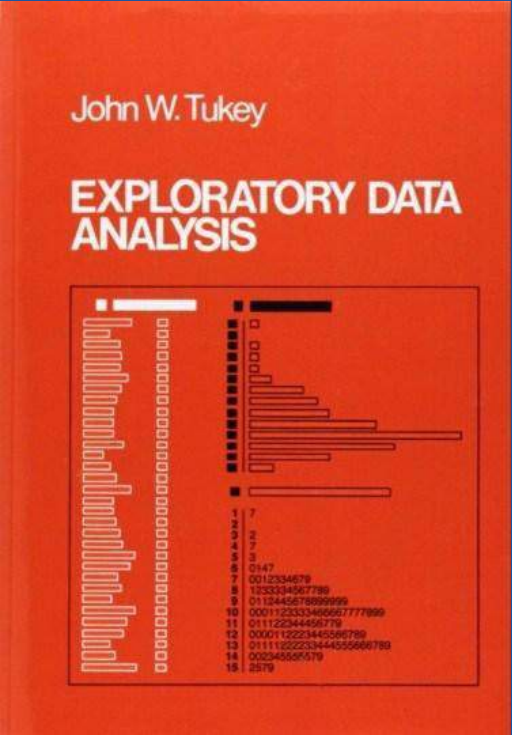
11:30 – Data analysis in action

12:00 – Qualitative data analysis with AI

12:30 - Close



Further reading



Activity 1

Can you tell me what the following dataset represents

<https://davetaz.github.io/number-cards>

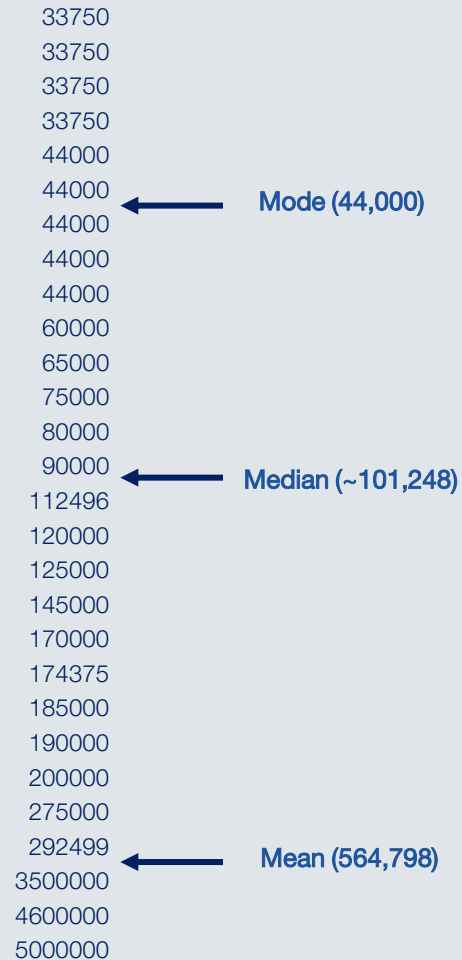
Clues

1. The dataset is complete
2. There is no rounding in the numbers
3. The numbers are discrete (none are additions of others)

Steps

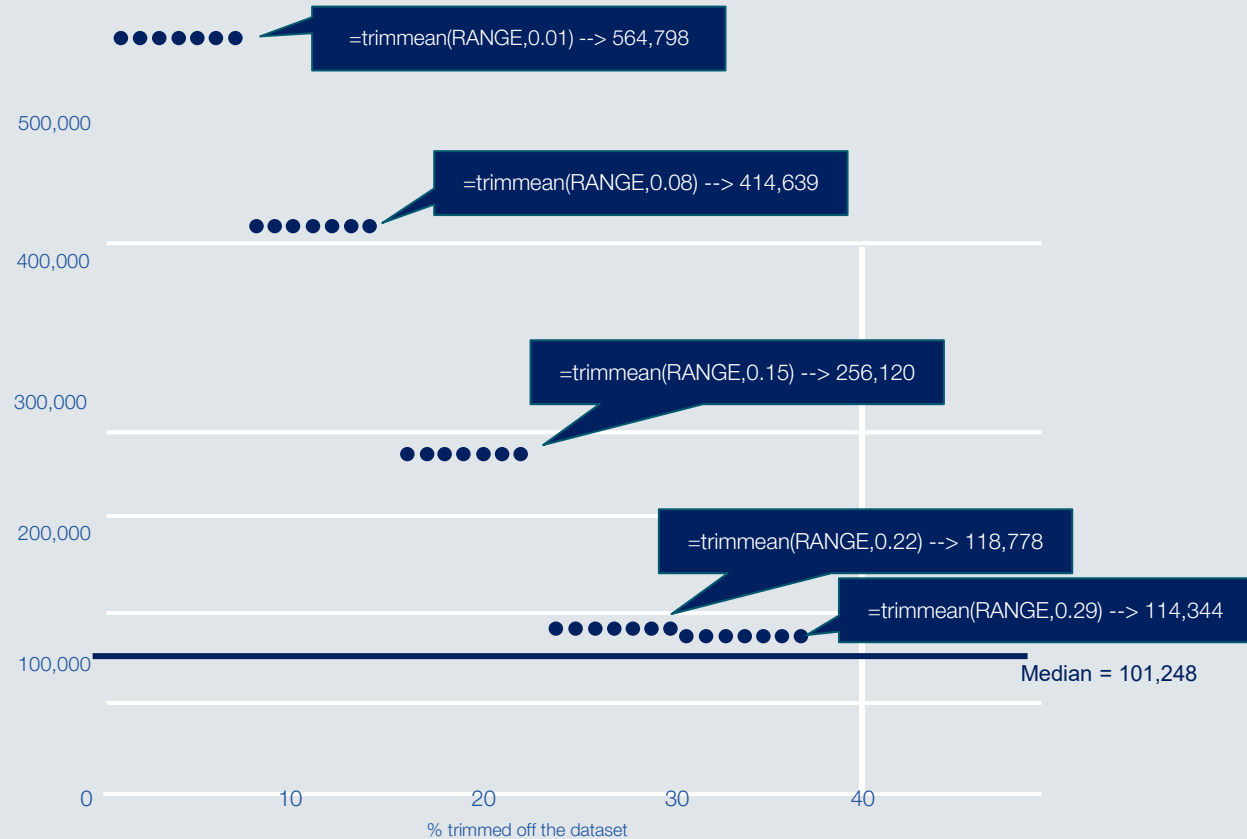
1. What facts can you tell me about the data?
2. If there were 28,000 datapoints not 28, what summary statistics would you use to give you an indication of the shape and insight from the data?
3. Given (1) and (2), what does this data represent?

Averages



Outliers

44000
44000
44000
44000
44000
60000
65000
75000
80000
90000
112496
120000
125000
145000
170000
174375
185000
190000
200000
275000

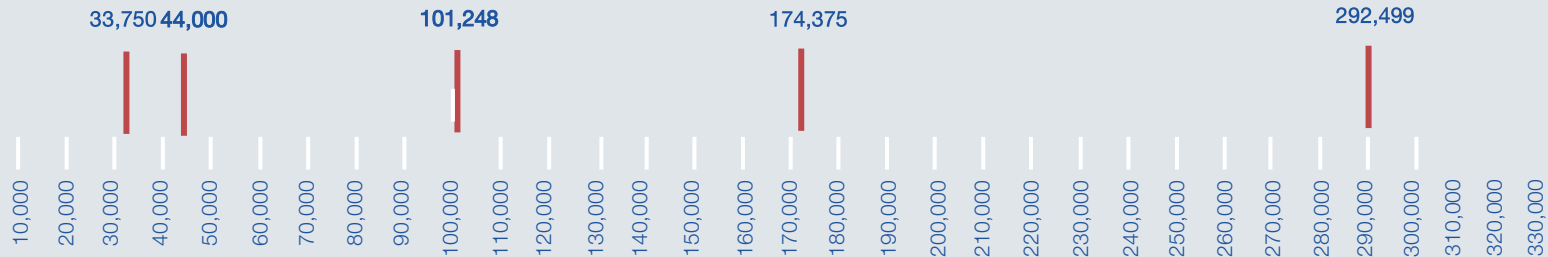


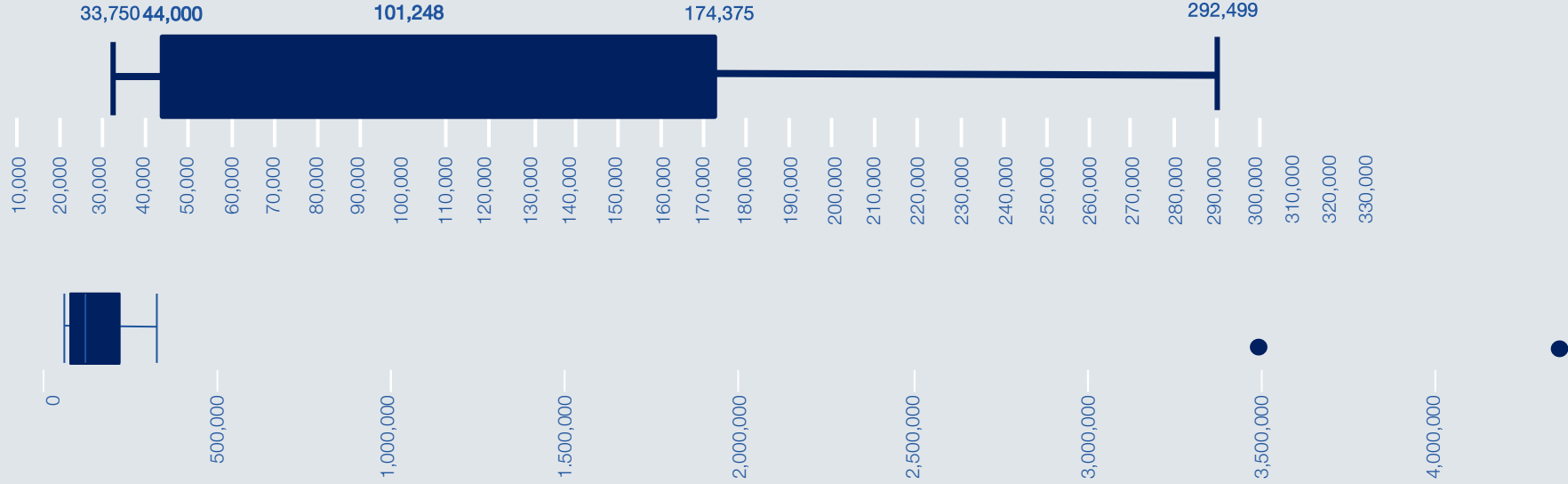
Formula to calculate a trimmed mean: `=trimmean(RANGE,percent)`

5-number summary: when mean or median isn't enough

33,750	33,750	33,750	33,750	44,000	44,000
44,000	44,000	44,000	60,000	65,000	75,000
80,000	90,000	112,496	120,000	125,000	145,000
170,000	174,375	185,000	190,000	200,000	275,000
292,499	3,500,000	4,600,000	5,000,000		

- Minimum value
- Maximum value
- Median
- Lower quartile
- Upper quartile





Shape of data

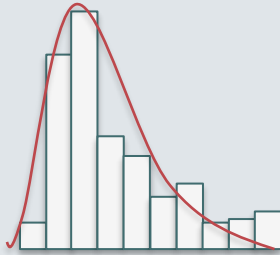
90,000
80,000
75,000
65,000
60,000
44,000
44,000
44,000
44,000
44,000
33,750
33,750
33,750

190,000
185,000
174,375
170,000
145,000
125,000
120,000
112,496

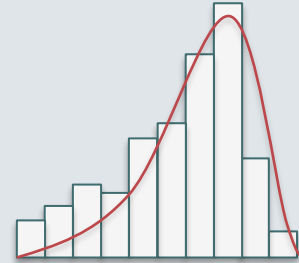
292,499
275,000
200,000

5,000
4,600
3,500

Distributions



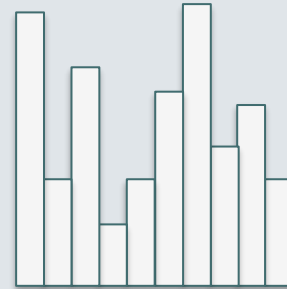
Positively Skewed
(mean is *skewed*
higher than median)



Negatively Skewed
(mean is *skewed*
lower than median)

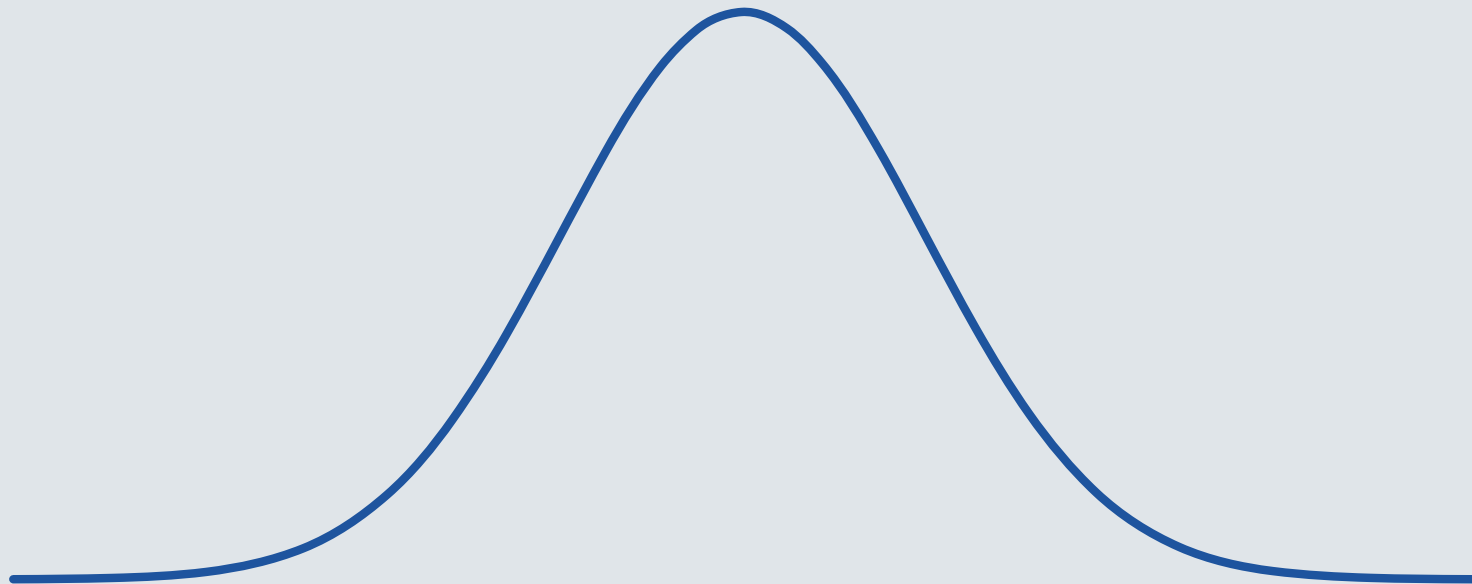


Normal distribution
(aka. Gaussian Distribution,
Bell Curve)

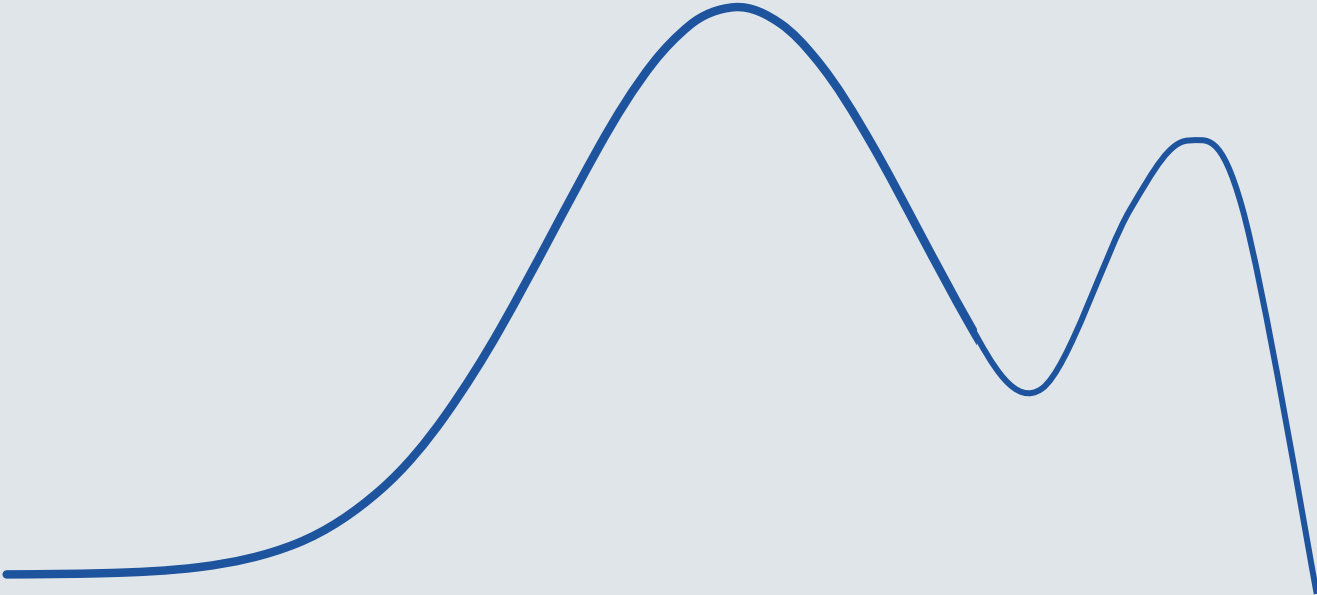


Multimodal distribution
(many modes)

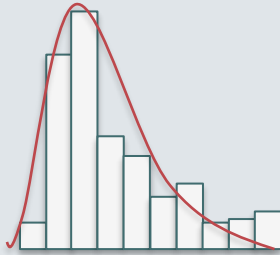
Histograms and targets



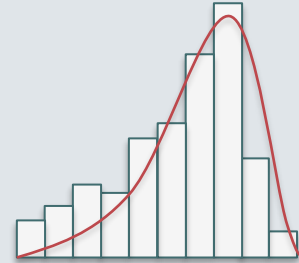
Histograms and targets



Distributions



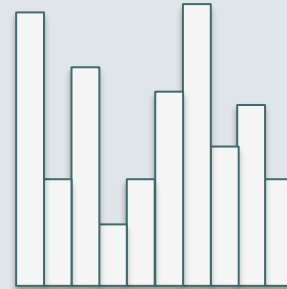
Positively Skewed
(mean is *skewed*
higher than median)



Negatively Skewed
(mean is *skewed*
lower than median)

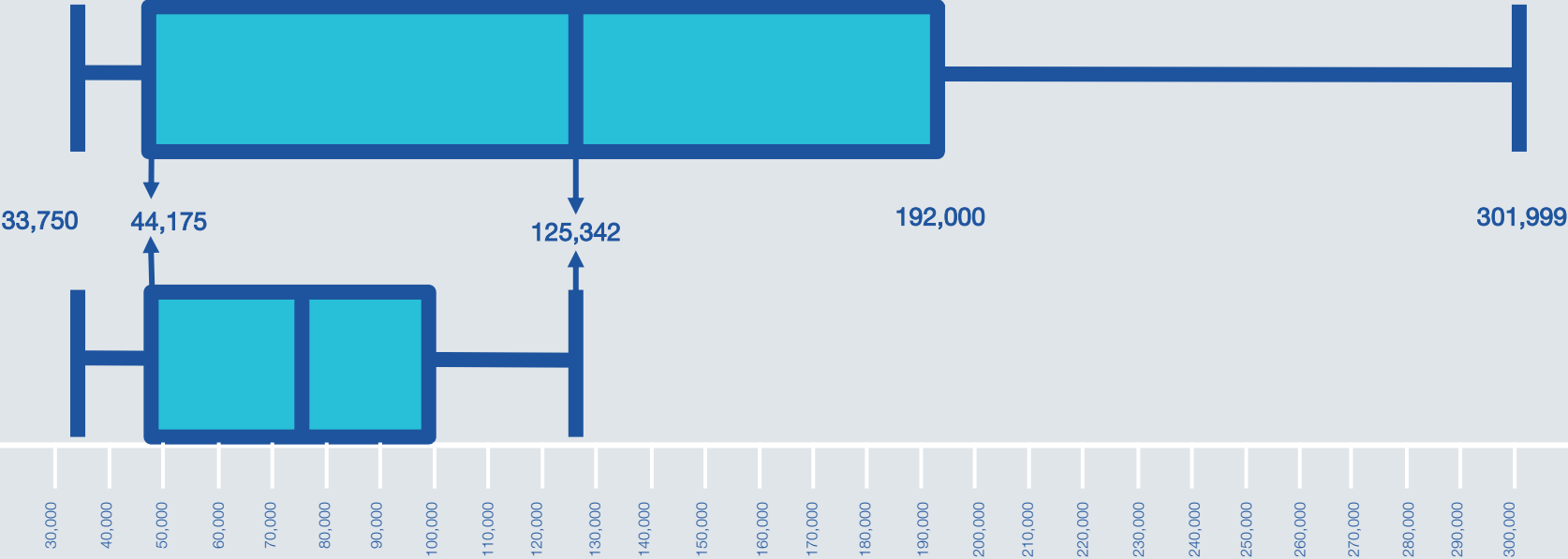


Normal distribution
(aka. Gaussian Distribution,
Bell Curve)



Multimodal distribution
(many modes)

Box plots are excellent for comparing datasets



NEWS

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Train strike: How much are rail workers paid?

7 December 2022

In a debate about the strikes in Parliament on 15 June, then Transport Secretary Grant Shapps said: "The median salary for a train driver is £59,000, compared with £31,000 for a nurse and £21,000 for a care worker."

The trouble with using the figure for train drivers in a debate about the RMT strikes is that the drivers are represented by their own union, Aslef, which is not taking part in the national strike.

To work out the median salary, if you put all train drivers in a row in order of their pay, the person in the middle of the row would be earning the median.

Mr Shapps was mostly right on these figures, which came from the Office for National Statistics (ONS).

Mr Shapps went on to say: "The median salary for the rail sector is £44,000, which is significantly above the median salary in the country."

We asked the Department for Transport (DfT) how it got to this figure and it initially said it had taken the median figures from the ONS for four categories of rail workers, added them up and divided by four:

This wouldn't have produced a median figure and the DfT subsequently got in touch to say they had actually got the figure from the ONS, which had produced a genuine median figure (£43,747) across everybody in those four categories, along with a fifth - workers who build and repair engines and carriages (their median salary was £46,753).

RMT general secretary Mick Lynch told BBC News on 6 December: "75% of the people in this dispute are paid less than £32,000 a year for working 24/7 shifts."

We asked the union how this had been worked out. It sent us a list dividing 33,000 members involved in the dispute into eight categories of workers, and giving the median salary for each of those categories.

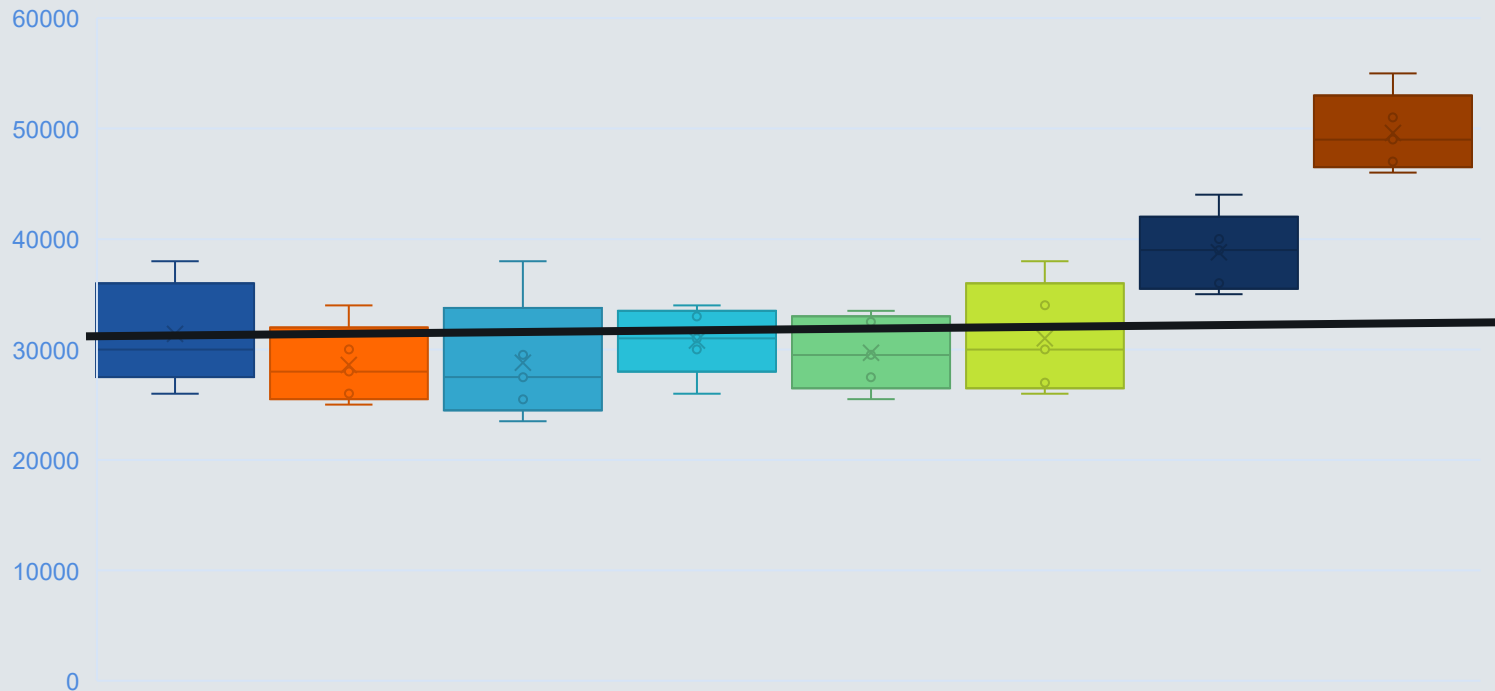
The list shows that three quarters of the workers are in categories for which the median salary is below £32,000, but that means some of them could be earning more than £32,000.

75% of workers are paid less than 32,000?

3/4 of Groups have medians less than 32,000

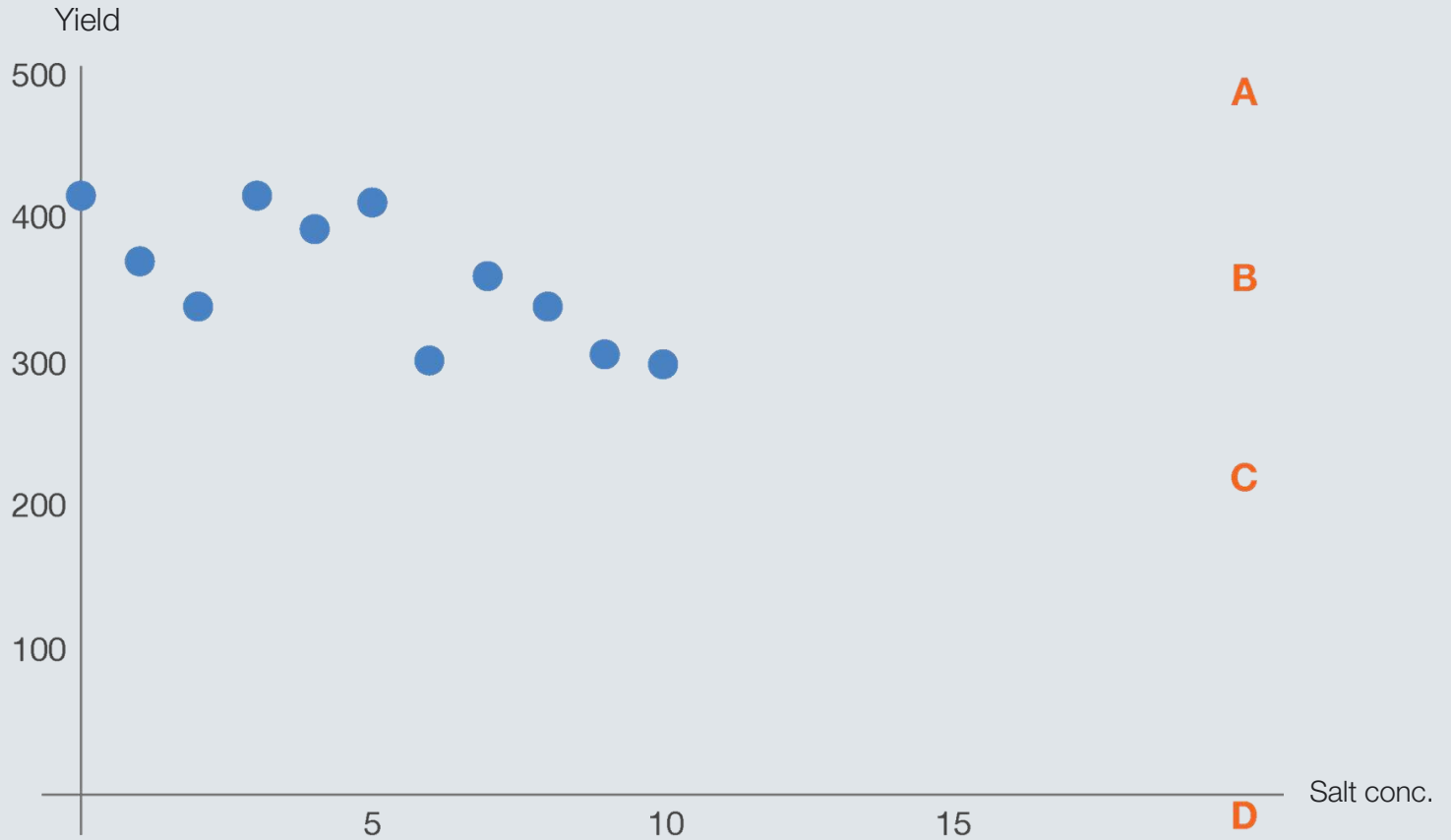
Is not the same as 3/4 of all data (it is only 44% of the quarters here)

Plus you cannot assume that each group contains the same number of workers

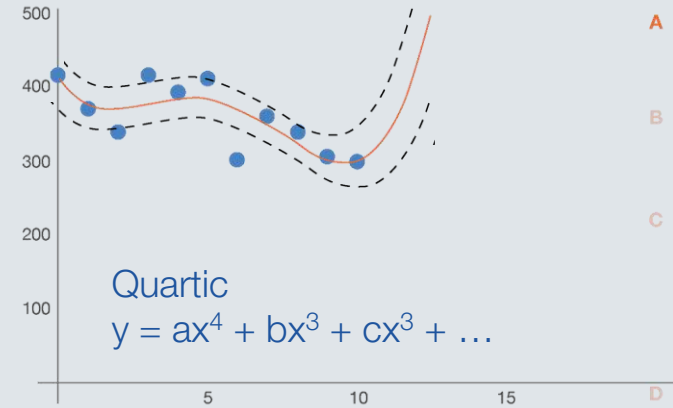
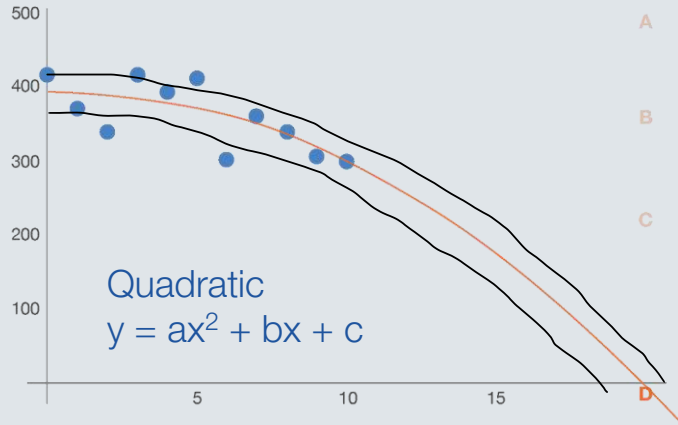
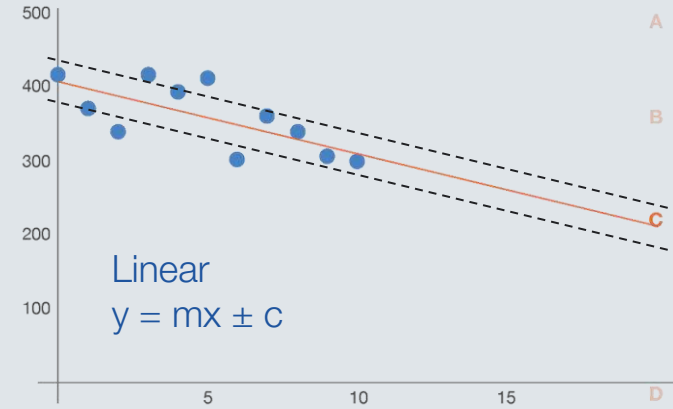
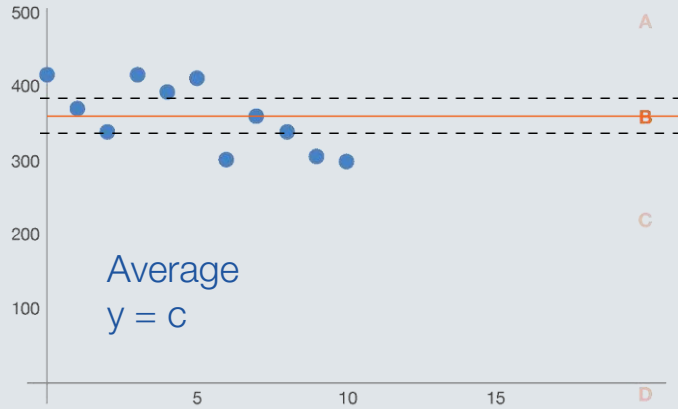


Inferential statistics allows you to make predictions and classifications based on data.

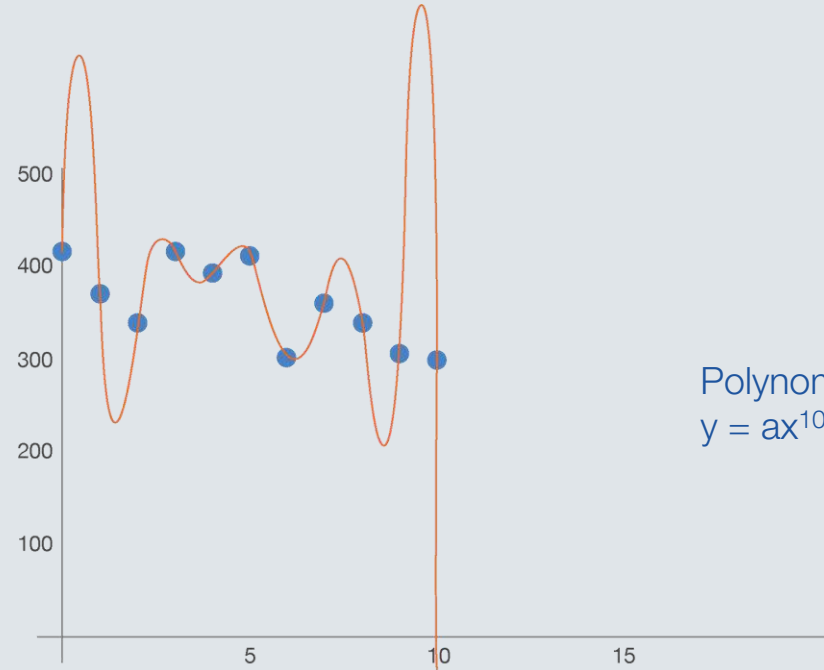
Trend Analysis



Overfitting



Overfitting



Polynomial
 $y = ax^{10} + bx^9 + cx^8 + \dots$

A perfect model of the past, is a rubbish predictor of the future!

Descriptive statistics vs Inferential statistics

* not only will it be rubbish you will open yourselves up to accusations of discrimination where it goes wrong.

Overfitting



Knowing which trade off is acceptable is key

Objective

To develop an image
recognition system that can
recognise cattle

True positive:

Algorithm says:
“Is Cattle”

is correct

False positive:

Algorithm says:
“Is Cattle”

is wrong

True negative:

Algorithm says:
“Not Cattle”

is correct

False negative:

Algorithm says:
“Not Cattle”

is wrong

Knowing which trade off is acceptable is key

Objective

To develop an AI system that
can diagnose cancer

True positive:

Algorithm says:
"Has cancer"

is **correct** - patient has
cancer

False positive:

Algorithm says:
"Has cancer"

is **wrong** - patient does not
has cancer

True negative:

Algorithm says:
"No cancer"

is **correct** - patient has not
got cancer

False negative:

Algorithm says:
"No cancer"

is **wrong** - patient has got
cancer

Calculating recidivism risk

Prisoners given questionnaire by trained prison staff.

Each question earns 1 point if negative.

Prisoners are determined to be high, medium or low risk for recidivism–this score is presented to a judge at sentencing.

Research found that only 11 of the questions had a statistical associate with recidivism.



Variable	%	Variable	%
Agreement		Agreement	
I. Criminal History		V. Accommodations	
1. Any prior convictions?	96	27. Unsatisfactory	64
2. Two or more prior convictions?	93	28. Three or more address changes last year?	82
3. Three or more convictions?	93	29. High crime neighborhood?	67
4. Three or more present offenses?	81	VI. Leisure/Recreation	
5. Arrested under age 16?	78	30. Absence of recent participation in an org. activity?	53
6. Ever incarcerated upon conviction?	95	31. Could make better use of time	53
7. Escape history from a correctional facility?	81	VII. Companions	
8. Ever punished for institutional misconduct?	87	32. A social isolate?	68
9. Charge made for probation/parole suspended during prior community supervision?	91	33. Some criminal acquaintances?	77
10. Official record of assault/violence?	86	34. Some criminal friends?	59
II. Education/Employment		35. Few anti-criminal acquaintances?	61
11. Currently employed?	86	36. Few anti-criminal friends?	59
12. Frequently unemployed?	72	VII. Alcohol/Drug Problem	
13. Never employed for a full year?	72	37. Alcohol problem, ever?	76
14. Ever fired?	78	38. Drug problem, ever?	88
15. Less than regular grade 10?	85	39. Alcohol problem, currently?	72
16. Less than regular grade 12?	88	40. Drug problem, currently?	68
17. Suspended or expelled at least once?	76	41. Law violations?	79
18. Participation/performance	78	42. Marital/family?	68
19. Peer interactions	76	43. School/work?	62
20. Authority interactions	75	44. Medical?	78
III. Financial		45. Other indicators?	63
21. Problems?	60	IX. Emotional/Personal	
22. Reliance upon social assistance?	69	46. Moderate interference?	84
IV. Family/Marital		47. Severe interference, active psychosis?	93
23. Dissatisfaction with marital or equivalent situation?	64	48. Mental health treatment, past?	87
24. Non-rewarding, parental	62	49. Mental health treatment, present?	89
25. Non-rewarding, other relative	65	50. Psychological assessment indicated?	66
26. Criminal family/spouse?	68	X. Attitudes/Orientation	
		51. Supportive of crime	68
		52. Unfavorable toward convention	78
		53. Poor, toward sentence?	72
		54. Poor, toward supervision?	62

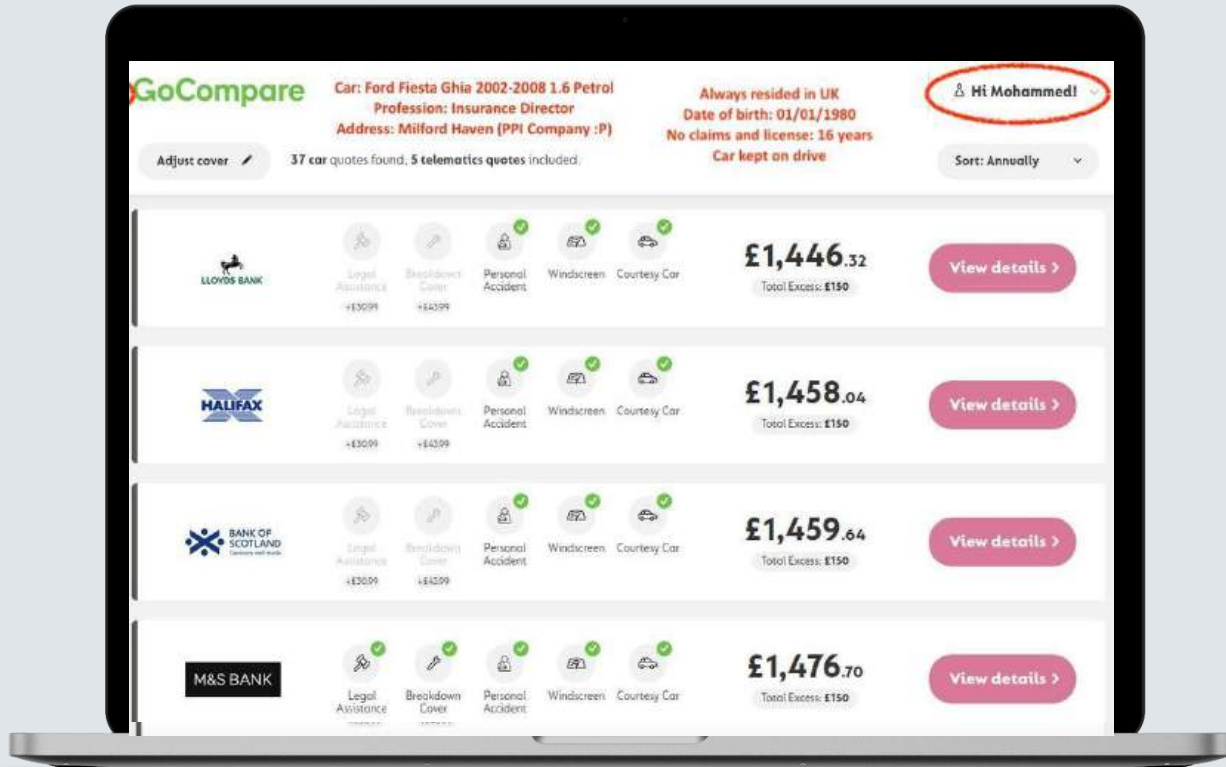
Car Insurance

Disclaimer: This story is from 2018. It was published in the Sun and Mirror newspapers.

The screenshots shown here were taken on 22 Jan 2018 as the story broke by David Tarrant to verify the story.

For the original story, please refer to the articles in the [Sun](#) and [BBC](#). Or for an update in 2023 please see [this report](#) from citizens advice.

These screenshots are for use ONLY by the Open Data Institute in training to highlight the stories findings. We are in no way accusing any organisation or any behaviour. We are highlighting the output that results from changing the input. We cannot see into the black-box or make assumptions as to why.



London fire brigade response times

In 2013, the London Fire and Emergency Planning Authority (LFEPA) adopted the Fifth London Safety Plan to reshape the service for 2014–2017.

As part of this plan, 10 fire stations were scheduled for closure on 9 January 2014, and 14 pumping appliances removed to reduce costs. ([London Fire Brigade](#))

The official rationale was to achieve budget savings of around £29 m (part of a broader £45 m target) while still maintaining response targets and keeping London safe. ([London Fire Brigade](#))

As the London Fire Brigade publish all their incident response data as open data we can do some analysis ourselves to find out what happened.



<https://london-fire.labs.theodi.org>

Qualitative data analysis and AI

Qualitative data analysis

Qualitative data is non-numerical, descriptive information that captures experiences, perceptions, reasoning, and context.

In government and organizational settings, this commonly includes:

- Interviews and focus groups
- Free-text survey responses
- Public consultation submissions
- Case notes, complaints, emails, and correspondence
- Reports, transcripts, and meeting notes
- Reviews and feedback (e.g. service or customer reviews)

Qualitative analysis helps us understand:

- Why people behave or respond in certain ways
- How policies, services, or decisions are experienced in practice
- What unintended consequences or tensions exist
- How impacts differ across communities or contexts
- It complements quantitative analysis by adding meaning, explanation, and human perspective.

Core activities

Across sectors, qualitative analysis usually involves:

1. Familiarisation with large volumes of text
2. Coding passages according to themes or concepts
3. Iteratively refining categories
4. Comparing themes across groups, places, or time
5. Synthesising findings into insights or narratives

This work is:

- Interpretive rather than mechanical
- Dependent on context and judgement
- Time-intensive and difficult to scale
- As a result, qualitative evidence is often under-used, even when it exists in abundance.

1

Become familiar with
data

2

Code into themes

3

Search for patterns
and connections

4

Synthesize findings

What generative AI can support

Large language models are particularly effective at working with language-rich datasets.

They can help:

- Rapidly synthesise large volumes of qualitative material
- Identify recurring themes and patterns
- Compare how issues are framed across datasets or groups
- Surface illustrative examples and quotations
- Support exploratory questioning and iteration
- This does not replace qualitative analysis — it augments the early and exploratory stages, freeing human analysts to focus on interpretation and decision-making.

Demo

1) [Download this dataset](#)

2) Upload it to your favourite AI tool and give it this prompt.

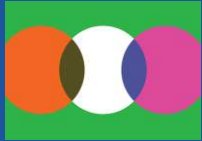
“Today we are going to do some qualitative analysis of users reviews for two restaurants contained in @reviews.csv.

What we are looking for is to summarise the reviews in several key areas.

- 1) Food
- 2) Drink
- 3) Service
- 4) Cost

Create a new file with the summative reviews in split into these areas of each. Do not make anything up.”

Strategy and Ethics



Data Ethics Essentials

Develop an understanding of data ethics as a domain.

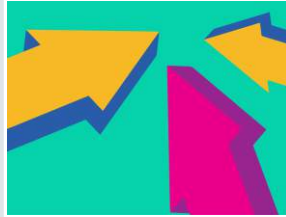
Explore the importance of data ethics practices through real-world examples.

Reflect on your own professional practice in relation to data and data ethics.

Apply tools and frameworks to support ethical thinking around the collection, sharing and use of data.

<https://learning.theodi.org/courses/data-ethics-essentials>

Cost: £FREE



Strategic Data Skills

This course will equip you with the theoretical knowledge, practical skills and technical know-how to make better decisions using data. It's a comprehensive course for people who want to work strategically with data, but don't need or want to learn complex programming.

Tutor Led

Self-Paced



Certified Data Ethics Professionals

Become a certified data ethics professional.

In this 60 hour course you will learn the knowledge, skills, and ability to apply tools and educate others on how to use, collect and share data responsibly, minimising harmful impacts. Modules include what is data ethics; ethics and personal data; working with data from 3rd party sources; data analysis and ethics in practice and operationalising data ethics.

Understand data ethics and AI (self-paced)



The Foundations of Ethical Decision Making

Explore different ethical frameworks, their strengths, their weaknesses and application to AI. Learn how to navigate ethical challenges through a principled step by step SAD-AM model. Explore the inescapable presence of bias and break the boundary between ethics and law. Ensure inclusivity, ethical growth and organisational impact.



Legal and Regulatory Frameworks

Learn about the pivotal regulations that exert influence across all phases of the data lifecycle, with a particular emphasis on emerging AI regulations. Understanding the legal and regulatory frameworks that govern data practices to ensure compliance. Understand the purpose of data licences, interoperability and the importance of transparency to maintain trust.



Ethics and Personal Data

Develop the skills to handle personal data responsibly. Investigate data anonymisation techniques: how to mitigate the risks whilst maintaining the utility of data. Gain hands-on experience in handling sensitive information and engage with our AI assistant tutor for instant feedback on your work.



Ethics in data collection

Explore ethical practices in the collection of data. Equip yourself with the knowledge and tools necessary to navigate the intricate terrain of data collection with ethics and integrity. Evaluate the sources of bias during data collection and mitigation strategies.



Bias and Fairness in Data and AI

Develop skill to ensure that algorithms that process data are “fair”. Develop your understanding of bias in data and algorithms and actively identify instances of bias in real-world datasets. Uncover the impact and consequences of bias in data and artificial intelligence.



Transparency and Explainability in Data

Explore key aspects of building trust in data practices and AI models. Implement techniques for developing AI models that are explainable and interpretable. Examine existing ethical guidelines and frameworks that promote transparency and explainability and how this can be embedded in your organisational context.



Tools and Frameworks in Evaluating Ethical Data Practices

Explore practical tools and frameworks to assess and enhance ethical data practices. Get hands-on experience with our Consequence and Risk Evaluation (CARE) tool and implement ethical data evaluation seamlessly into your workflows. Address the possibility of unintended consequences arising from AI deployment.

Thank you

Course materials:

<https://training.theodi.org/ML101>



For more information about how the ODI can support your data need:

- Consulting services
- Research and policy
- Membership
- Training

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